MD Analysis Pipeline

User Manual

Automated GROMACS MD Analysis - Installation, Configuration, and Usage

Files included

File	Purpose
md_analysis.py	Main analysis script - never needs to be edited
md_config.ini	Configuration file - edit for each simulation
install.py	One-time dependency installer
USER_MANUAL.pdf	This document

Quick Start

Step	Command
1. Install dependencies (once)	python install.py
2. Edit md_config.ini	Set sim_dir = /path/to/your/sim/folder
3. Run analysis	python md_analysis.py /path/to/sim/folder

1. Install Dependencies

Run this once on any new computer before your first analysis. It installs all required Python packages automatically.

```
python install.py
```

Package	Used for
NumPy	Numerical calculations
Pandas	Data tables and CSV export
Matplotlib	Plotting
Seaborn	Plot styling
MDAnalysis	Reading trajectories - RMSD, RMSF, H-bonds, contacts
fpdf2	PDF generation (this manual)

GROMACS must also be available on your system. On HPC clusters, load it before running the analysis:

```
module load gromacs
```

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2. Configure md_config.ini

Open md_config.ini in any text editor (Notepad on Windows, TextEdit on Mac, gedit or nano on Linux). The only required change for each simulation is sim_dir.

Required - set the simulation folder

Change sim_dir and out_dir to point at your GROMACS files:

```
[paths]
sim_dir = /home/user/projects/1IEP_MD
out_dir = /home/user/projects/1IEP_MD/results
```

Use the full absolute path. out_dir is created automatically if it does not exist.

File names are auto-detected

The script scans sim_dir and picks .tpr, .xtc, .edr, .gro, and .ndx files automatically, preferring files whose name contains 'md' (skipping em, npt, nvt). Uncomment and set file names manually only if detection picks the wrong file:

```
[paths]
tpr = md_production.tpr
xtc = md_production.xtc
```

Common optional settings

Change the time axis to nanoseconds

```
[analysis]
time_unit = ns
```

Add a label to all plot titles

```
[system]
sim_label = ABL1 WT + Imatinib 100 ns
```

Set the gatekeeper residue (0 = skip)

```
[system]
gatekeeper_res = 315
```

Set ligand residue name (if auto-detection is wrong)

```
[system]
lig_resname = LIG
```

Use a different GROMACS command (e.g. on HPC clusters)

```
[system]
gromacs_cmd = gmx_mpi
```

Adjust plot appearance

```
[figure]
dpi            = 300      # 72=screen  150=standard  300=publication
width          = 12
height         = 5
title_fontsize = 14
label_fontsize = 12
ligand_color   = #ff6600
time_unit      = ns
```

3. Run the Analysis

Pass the simulation folder as an argument. The script looks for md_config.ini inside that folder automatically:

```
python md_analysis.py /path/to/1IEP_MD
```

Or pass the config file directly:

```
python md_analysis.py --config /path/to/1IEP_MD/md_config.ini
```

Running multiple simulations

Keep one copy of md_analysis.py. Place a copy of md_config.ini in each simulation folder and change only sim_dir in each one:

```
python md_analysis.py /data/1IEP_MD
python md_analysis.py /data/3IK3_MD
python md_analysis.py /data/T315I_MD
```

Each run writes its results to its own out_dir, so the outputs never mix.

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4. What the Script Does

The pipeline runs ten steps in order. Each step is independent - if one fails, the rest continue.

Step 1 - PBC Correction

Fixes periodic-boundary artefacts using GROMACS trjconv. Produces a corrected trajectory (md_center.xtc) used by all subsequent steps. Skipped automatically if the corrected trajectory already exists.

Step 2 - Thermodynamic Properties

Extracts potential energy, kinetic energy, total energy, temperature, pressure, volume, and density from the .edr file using gmx energy. Each property is saved as an individual plot with mean +/- 1 sigma shading.

Step 3 - RMSD

Root-mean-square deviation of the protein backbone Ca atoms (structural stability) and the ligand heavy atoms fitted on the backbone (binding pose stability). Values consistently below 2-3 Angstroms indicate a stable simulation.

Step 4 - RMSF

Per-residue Ca root-mean-square fluctuation. Shows which parts of the protein are flexible. The gatekeeper residue is highlighted with a vertical marker.

Step 5 - Radius of Gyration

Protein compactness over time using gmx gyrate. A stable Rg indicates the protein maintains its folded conformation.

Step 6 - Protein-Ligand H-bonds

Counts hydrogen bonds between protein and ligand at every frame. Reports per-pair occupancy (% of frames each H-bond is present).

Step 7 - Gatekeeper Distance

Minimum distance between the ligand and the gatekeeper residue (e.g. THR315 in ABL1) over the trajectory. Shows whether the ligand maintains contact with this key residue throughout the simulation.

Step 8 - Binding-Pocket Contact Map

Identifies all protein residues within the contact cutoff (default 4.5 A) of the ligand and reports their occupancy as a bar chart. Residues below min_plot_contact_pct are hidden.

Step 9 - Summary Figure

A four-panel figure combining backbone RMSD, ligand RMSD, radius of gyration, and per-residue RMSF in one image.

Step 10 - Text Report

A plain-text summary of all key numerical results saved to tables/analysis_report.txt.

5. Output Files

All output is written to out_dir (default: results/ inside the simulation folder).

File	Contents
energy_potential.png	Potential energy vs time
energy_kinetic.png	Kinetic energy vs time
energy_total_energy.png	Total energy vs time
energy_temperature.png	Temperature vs time
energy_pressure.png	Pressure vs time
energy_volume.png	Volume vs time
energy_density.png	Density vs time
rmsd.png	Backbone and ligand RMSD (two panels)
rmsf.png	Per-residue Ca RMSF
gyration.png	Radius of gyration
hbonds.png	H-bond count vs time
gatekeeper_distance.png	Ligand-gatekeeper minimum distance
pocket_contacts.png	Binding-pocket contact occupancy bar chart
summary_panel.png	Four-panel summary figure
tables/rmsd.csv	RMSD data (time, backbone, ligand)
tables/rmsf.csv	RMSF data (resid, resname, rmsf_A)
tables/hbonds_count.csv	H-bond count per frame
tables/hbonds_pairs.csv	Per-pair H-bond occupancy table
tables/gatekeeper_distance.csv	Gatekeeper min-distance per frame
tables/pocket_contacts.csv	Contact occupancy per residue
tables/analysis_report.txt	Full numerical summary report

6. Troubleshooting

GROMACS not found

Either GROMACS is not installed or not on your PATH.
On HPC: run `module load gromacs` before the analysis.
Custom name: set `gromacs_cmd = gmx_mpi` in [system] of md_config.ini.

Missing files in sim_dir

Check that sim_dir is the correct folder. The script prints what it found.
If your file names are non-standard, set them explicitly in [paths].

Multiple non-protein residues found

The script used the first one. If wrong, set the correct residue name:

```
[system]
lig_resname = LIG
```

Protein_LIG group NOT FOUND

The script tries to create this group automatically with gmx make_ndx.
If that also fails, find the correct group number in the startup table and add:

```
[groups]
Protein_LIG = 22
```

MDAnalysis steps are skipped

MDAnalysis is not installed. Run `python install.py` to fix this.

Plots show 'No data'

That step's analysis failed silently. Check the terminal output for a [WARN] message on that step for the specific error.

7. Quick-Reference - All Settings

Section	Key	Default	Description
[paths]	sim_dir	(required)	Folder with GROMACS files
[paths]	out_dir	results	Output folder (auto-created)
[system]	sim_label	simulation	Label in plot titles
[system]	gromacs_cmd	gmx	GROMACS executable name
[system]	lig_resname	auto	Ligand residue name
[system]	gatekeeper_res	315	Gatekeeper residue (0=skip)
[analysis]	time_unit	ps	Time axis unit: ps or ns
[analysis]	contact_cutoff_A	4.5	Contact distance cutoff (A)
[analysis]	hbond_distance_A	3.5	H-bond donor-acceptor cutoff (A)
[analysis]	hbond_angle_deg	120	H-bond angle cutoff (degrees)
[analysis]	min_plot_contact_pct	10	Min occupancy % shown in contact map
[figure]	dpi	150	Image resolution (DPI)
[figure]	width / height	10 / 4	Figure size in inches
[figure]	title_fontsize	13	Plot title font size (pt)
[figure]	label_fontsize	11	Axis label font size (pt)
[figure]	font_scale	1.1	Overall text scale multiplier
[figure]	backbone_color	#2c7bb6	Protein backbone trace colour
[figure]	ligand_color	#d7191c	Ligand trace colour
[figure]	hbond_color	#1a9641	H-bond trace colour
[figure]	gatekeeper_color	#fd8d3c	Gatekeeper trace colour
[figure]	rmsf_color	#756bb1	RMSF trace colour
[figure]	energy_color	#636363	Energy trace colour
[figure]	ligand_marker_size	60	Marker size on RMSF plot (pt2)